Nebulae instruments: what is on the stick, and how to choose

The guides in this manual cover the instruments we made: Pure Data (gloop, glitch, spectral, topo, shimmer, resonator, chop, vocoder), SuperCollider (drift, strike, cloud), Csound (Pearls, stretch) and the Bluetooth storyteller. The other instruments on the stick have their own manuals (listed below).

Choosing an instrument (custom firmware uf2)

1. **Hold Speed for 1.5 s**: the selector opens.
2. **Turn Speed** to pick the bank: anticlockwise = **Csound user** (left light), middle = **factory** (both lights), clockwise = **Pd** (right light green), clockwise again = **SuperCollider** (right light **blue**).
3. **Turn Pitch** to pick the page of five. The Pitch light shows the page: white = 1–5, orange = 6–10, yellow = 11–15, blue = 16–20, green = 21–25.
4. **Lit buttons** have an instrument; the bright one is running. **Press one** and it loads straight away.
5. Pitch or Speed **click** leaves without loading.

The five buttons, left to right, are **Record, File, Source, Reset, Freeze**: positions 1–5 of each page.

The module sorts the files by name (digits before letters), so a numeric prefix sets the position. Pd and SC files starting with `_` are hidden. `nebulae_banks.txt` on the stick is this map, regenerated from the files (`tools/stick_manifest.py`).

The stick, 2026-11-13

Bank / page	Button	Instrument	What it is
Csound user, white	Record	001_a_granular_looper_Factory_Default	An older release of Qu-Bit's granular looper (its overdub buffers differ from the built-in one)
	File	002_avalon	AVALON "Morphing Nebulaeum": an 8×8×64 wavetable morph oscillator after the Synthesis Technology Terrariums; reads WaveEdit banks (`ROM A/B/C.wav`, `avalon.wav`). Manual: `avalon.pdf`, `avalon.txt`
	Source	003_bowed_drone	Bowed Drone (Collin Russell; V1 port by Chris Erin): evolving bowed-string drones. Diagram: `Bowed_Drone_Diagram.pdf`
	Reset	004_mars_affects	Mars Affects 1.5.1: filter, VCA, delay, effect, reverb and ducking in a chain (after the factory Effects / World of Echo). Manual: `mars_affects.pdf`
	Freeze	005_realms_v1.0	Realms (Robert Pabst, Ian Hoogeboom): morph through up to seven "rooms" of paired ambient loops (the `11_…72_` .flac files). Manual: `realms.pdf`, `realms.txt`
Csound user, orange	Record	006_stretch	stretch: extreme time-stretch and freeze of stick files or of what you just played (its guide is in this manual)
	File	999_fw_backup	Utility: backs up the firmware and writes an inventory to the stick (see Utilities)
	Source	bt_storyteller	The granular looper, telling your

Bank / page	Button	Instrument	What it is
			PC what happens to it over Bluetooth (its guide is in this manual)
	Reset	bt_test	Utility: the Bluetooth check (see Utilities)
	Freeze	pearls	Pearls: Mutable Instruments Beads (its guide is in this manual)
Factory (built in)	1–5	a_granularlooper, b_oneshot, c_synth, d_effects, e_testio	Qu-Bit's own: see the Nebulae manual
Pd, white	Record	A_durm	No notes on the stick. From its patch: a drum processor with side-chain compression, trigger glitches, filter and downsampler; holding Source turns the burst off
	File	B_Plasma_Membrane	No notes on the stick. Its knobs each control two parameters (split down the middle)
	Source	C_ReconetC	Reconetc: a generative dual sample slicer with four micro-algorithms in zones, trigger probability, repeats and two reverbs. Needs its six samples (`Atmos`, `Kick`, `Piano1`, `Piano2`, `SnareCrickets`, `Stabs` .wav). Notes: `Reconetc redme.txt`
	Reset	D_rhythmic_chords	Qu-Bit's Pd example (also in the firmware): chords on V/oct with a rhythm
	Freeze	E_echo	Echo: the 4-head looper (the "gloop" guide in this manual; named Echo on the stick so it isn't taken for the Cutlasses Gloop)
Pd, orange	Record	F_glitch	glitch: clocked beat-repeater
	File	G_spectral	spectral: Spectral Clouds+
	Source	H_topo	topo: drum patterns + drum synth
	Reset	I_shimmer	shimmer: shimmer reverb
	Freeze	J_resonator	resonator: exciter → resonator → delay
Pd, yellow	Record	K_chop	chop: slice sequencer (its guide is in this manual). *To copy: not on the stick yet*
	File	L_vocoder	vocoder: 20-band vocoder (its guide is in this manual). *To copy: not on the stick yet*
	Source	ZZ_pitchlog	Utility: measures the Pitch control's timing (see Utilities)
SuperCollider, white	Record	01_sctest	Test patch: a plucked saw for checking the controls
	File	02_sc_drift	drift: drone / pad
	Source	03_sc_strike	strike: plucked string
	Reset	04_sc_cloud	cloud: granular input
	Freeze	31-simple	Technobear's example: a saw on Pitch. **Loud**: turn the output down first

gloop — 4-head looper for the Nebulae (user guide)

A Pure Data take on the Cutlasses **Gloop** looper. Record a loop (up to 2 minutes, stereo), then play it back with **four play heads** at once. Each head can play a different part of the loop, at its own speed, direction, pitch, volume and pan. That makes it good for layered loops, phasing, canons and granular-style textures.

Panel map

```
                [ speed ]                          speed  = RATE of selected head
 start                          size               start  = POSITION of selected head    (alt: VOLUME head 1)
                                                   size   = SIZE of selected head        (alt: VOLUME head 2)
 density       [ pitch ]        overlap            density= JITTER of selected head      (alt: VOLUME head 3)
                                                   overlap= PAN of selected head         (alt: VOLUME head 4)
         blend          window                     pitch  = TRANSPOSE ALL heads, 1V/oct  (alt: INTERVAL of selected head)
                                                   blend  = input <-> loop mix           (alt: DEGRADE)
                                                    window = CROSSFADE at loop edges      (alt: TAPE WOBBLE)

[record]        [file]           [source]          [reset]           [freeze]
REC/Overdub     PAUSE            HEAD bit1         RESTART           HEAD bit2       (hold Source = alt layer)
alt:CLEAR       alt:GRID
```

Choosing a head (watch the Source and Freeze LEDs)

Head	Source LED	Freeze LED
1	off	off
2	ON	off
3	off	ON
4	ON	ON

Tap Source: swaps 1↔2 and 3↔4.
Press Freeze: swaps 1↔3 and 2↔4.
The LEDs always show the selected head.

Gates into the Source and Freeze jacks select heads too.

Knobs (normal layer: edit for the selected head)

Knob	Does	Notes
start	where the head's sub-loop sits in the main loop	
size	length of the head's sub-loop	from the whole loop down to tiny grain-sized slices
density	jitter: how far the sub-loop jumps randomly on each repeat	0 = steady
overlap	pan	CCW = left, centre = both, CW = right
speed (encoder)	playback rate and direction	centre = stopped. Click = 1× forward, click again = 1× reverse. Turn out to 4× either way
pitch (encoder + 1V/oct jack)	transpose **all heads** together, in semitones	click = original pitch. Heads keep their intervals, so a V/oct sequence plays the whole chord of heads in tune
window	crossfade at the sub-loop edges (all heads)	CCW = 5 ms (tight) … CW = 1 s (soft, swelling)
blend	dry input ↔ loop playback	CW = loops only

Takeover (start, size, density, overlap, speed): when you switch heads, a knob doesn't change the new head until you **turn** it. Each head keeps its own settings until you deliberately change them. CV into a knob counts as turning it, so CV always modulates the **currently selected** head.

Alt layer (holding Source)

Control	Does
start / size / density / overlap	volume of heads **1 / 2 / 3 / 4** respectively
blend	degrade: each pass gets quieter and duller (sound-on-sound fade)
window	tape wobble (wow and flutter) on all heads
pitch (encoder)	**interval of the selected head**: semitones above/below the global pitch, ±2 octaves. The Pitch LEDs show the amount. Set per head (select the head, hold Source, turn)
Record	**clear the loop**
File	grid: off → 4 → 8 → 16 → off. Positions and sizes snap to 1/4, 1/8 or 1/16 of the loop

Heads 2–4 start **muted**. Hold Source and turn up **size**, **density** or **overlap** to hear them.

> **Never click the Pitch encoder while holding Source.** On the Nebulae that resets every alt setting to default: the head volumes jump, and the encoder's alt value resets. The head intervals are protected (the patch ignores the jump), but you'll need to turn the encoder back up before adjusting an interval again.

Buttons (normal layer)

Button	Does
Record (+gate)	1st press: start recording. 2nd press: close the loop and start playing. Later presses: overdub on/off
Reset (+gate)	restart all heads from the start of their sub-loops
File	pause the selected head (it fades and winds down like tape). Press again: it restarts from the top
Source / Freeze	select the head (see above)

Quick start

1. Patch audio into **In L** (and **In R** for stereo). Set **blend** to about noon.
2. Press **Record**, play something, press **Record** again. The loop plays on head 1.
3. Turn **blend** fully CW to hear only the loop.
4. **Tap Source** (Source LED on = head 2). Hold Source and turn **size** up to hear head 2.
5. Let go of Source. Turn **speed** a little clockwise (head 2 now runs faster than head 1) and turn **size** down. Head 2 now loops a shorter, faster slice against head 1.
6. **Press Freeze** (both LEDs on = head 4). Hold Source, raise **overlap** for its volume and turn **pitch** down for an octave below. Let go and set its **speed** to reverse.
7. Patch a sequencer into **V/oct**: every head follows, keeping the intervals you set.
8. Press **Reset** to bring all heads back into sync.
9. Hold Source + **Record** to clear and start again.

Tips

- **Granular:** small **size**, some **density** (jitter), **window** CW for soft grains, several heads at different intervals.
- **Chords:** give heads 2–4 intervals (e.g. +7, +12, −12) with Source + Pitch, then play the chord from V/oct.
- **Rhythmic slicing:** hold Source and press **File** to set grid 8 or 16, so sizes and positions lock to the beat.
- **Sound on sound:** hold Source and raise **blend** (degrade), then turn overdub on. Older layers slowly fade away.
- **Tape vibe:** a little alt **window** (wobble) and slow **speed** changes.
- Gate into **Reset** to retrigger all heads in time with a clock.

Limits (compared with the real Gloop)

- Not yet included: delay, reverb, bit crusher, wavefolder, tape saturation/hiss/dropouts, tilt EQ.
- No one-shot or jump triggers, overdub only (no replace mode).
- Long crossfades dip in volume at the loop point rather than overlapping.

glitch — clock-synced beat-repeater for the Nebulae (user guide)

A Pure Data port of **Glitch** (Music Thing Workshop Computer program 57). The Nebulae is always recording into a 43-second circular buffer. Each clock pulse marks one **beat**, and the patch plays that beat back in one of two modes:

- **Glitch** — ratchets the start of the last beat (÷1 to ÷6). Each repeat may be reversed, and may be degraded with sample-rate reduction and bitcrushing.
- **Stutter** — a live breakbeat slicer. It cuts the beat into 1–8 slices and plays them in a shuffled order. Slices may repeat, and they may come from earlier beats.

When a slice isn't glitched, live audio passes straight through, so the effect drops in and out of your signal.

Panel map

```
              [ speed ]                  start   = GATE: above half = glitch every slice
 start                         size      size    = DIVISION  Glitch ÷1 ÷2 ÷3 ÷4 ÷6 / Stutter 1 2 3 4 8 slices
                                         density = PROBABILITY  Glitch: glitch + reverse / Stutter: drift
 density      [ pitch ]        overlap   window  = X  Glitch: degrade amount / Stutter: shuffle probability
                                         overlap = Y  Glitch: degrade probability / Stutter: repeat probability
        blend          window            blend   = dry <-> glitched
                                         pitch   = TRANSPOSE the glitches, 1V/oct (click = original)
                                         speed   = RAMP: each repeat steps up (right) / down (left), centre = flat

[record]  [file]  [source]  [reset]  [freeze]
 FORCE      -      MODE      CLOCK    FREEZE
```

Button / jack	Does	LED
Reset (jack or button)	**Clock in.** Each pulse = one beat; the time between pulses sets the beat length (up to 10 s). Tap the button twice to set a tempo by hand	flashes on each pulse
Source	**Mode:** off = **Glitch**, on = **Stutter**. Takes effect at the next clock	on = Stutter
Record	**Force:** glitch / shuffle every slice (the WC switch held Down). A gate into the Record jack toggles it	on = forced
Freeze	**Freeze** the buffer: recording stops, playback keeps glitching the frozen material. A gate into the Freeze jack toggles it	on = frozen

All six pots have CV jacks, so every parameter can be modulated.

Which slices get the effect

For every slice, in this order:

1. **Record LED on** → always (Force).
2. **Start above half** (pot, or a gate into the Start CV jack with the pot fully CCW) → always (gate).
3. Otherwise **random**, with the probability set by **density** (Glitch) or **window** (Stutter).

Anything else is live pass-through. For pure gate behaviour (the WC's middle switch position), set the probability to 0.

Glitch mode (Source LED off)

At each clock pulse the patch measures the beat. It then plays the **first 1/div of the previous beat**, repeated div times across the next beat.

Control	Does
size	ratchet division: ÷1 (repeat the whole beat), ÷2, ÷3, ÷4, ÷6 — five equal zones
density	glitch probability (switch-Up behaviour) **and** reverse probability, rolled per slice
window	degrade amount. CCW = clean. First half: sample-rate reduction up to 16×. Second half: 16× plus bitcrush, up to 7 bits removed
overlap	degrade probability per slice. CCW = never, CW = always

Stutter mode (Source LED on)

At each clock pulse the beat is cut into slices, which are reshuffled and played back across the next beat.

Control	Does
size	slices per beat: 1, 2, 3, 4, 8
density	**drift** probability: how often the beat is taken from an earlier beat (up to 12 beats back) instead of the last one. The slice grid stays locked to real beats
window	shuffle probability per slice (else live pass-through)
overlap	repeat probability: how often a slice plays again before moving on. CW = one slice stutters for the whole beat

Pitch and Speed (both modes)

The rhythm never changes; only the material inside each slice does.

Control	Does
pitch (encoder + 1V/oct jack)	**Transpose** in semitones. Click = original pitch; the range is −3 to +2 octaves. Live, so a sequencer into the 1V/oct jack plays the repeats as a melody. Glitched slices go up or down. **Pass-through goes down only** (see below)
speed (encoder + CV)	**Ramp** across the repeats of each beat: every slice after the first steps by N semitones. Left of centre = falling (a tape-slowing feel): left click = −3 semitones per repeat, left end = −12. **Centre to the right click = flat.** Past the right click = rising, up to +12 at the right end

A pitched-**up** slice plays its own material faster and then goes quiet until the next slice, so it sounds tighter and more staccato. A pitched-**down** slice plays only the start of its material, slower.

Pass-through with Pitch below the click is slice-locked. At every slice boundary it picks up the live input exactly where it is and plays it slower and lower, until the next boundary snaps it back. A hit on the beat grid comes out **in time, with no added latency**: drums stay locked to the clock, each hit sounds like tape slowing under it, and only hits between boundaries drift later. Use ÷4 or ÷8 for tighter re-syncs.

Pitch above the click: pass-through stays untransposed. Live audio can't be played faster than it arrives without adding delay.

While frozen: pass-through is plain live audio.

Speed's ramp applies to glitched slices only.

The total shift is limited to −48…+24 semitones, so steep ramps on ÷6 stop climbing at +2 octaves.

Speed remembers its last position. On the right click the repeats are plain. On the left click they fall by 3 semitones per repeat, so click it over for plain repeats.

Freeze

Press Freeze, or send a gate to its jack. The buffer stops recording and the clock keeps running, so you can keep re-cutting the same snapshot with any knob. Press it again to resume recording. The patch keeps playing the frozen beat until one full beat of new audio has been recorded, so there's no stale audio at the changeover.

Quick start

1. Patch a clock (one pulse per beat) into **Reset** and audio into the input(s) (stereo/mono). Set **blend** fully CW.
2. Turn **Record** on (Force): every beat now repeats the one before.
3. Turn **size** up through the zones for ÷2, ÷3, ÷4, ÷6 ratchets.
4. Turn Record off and raise **density** for random glitches and reverses.
5. Bring up **window** and **overlap** for lo-fi degradation.
6. Tap **Source** for Stutter mode: size fully CW (8 slices), Record on, and the beat is scrambled.
7. Press **Freeze** to hold a moment and keep chopping it.

Differences from the Workshop Computer version

- **Buffer:** 43 s instead of 2.33 s. Beats can be up to 10s long. Stutter drift reaches back up to 12 beats.
- **Outputs:** stereo processed audio only. Blend replaces the separate dry output.
- **Switch:** Up / Mid / Down becomes probability / Start gate / Record (Force), and they can be combined.
- **Freeze and Force** toggle on a gate's rising edge instead of following it, because those Nebulae inputs latch.
- **Mode** is the Source button and can change while playing.

Clock jitter: messages reach with about 1–3 ms jitter. Beat-length changes under 5 ms are ignored, and each beat's slices end about 3 ms early (plus a 30 ms grace period) so a late pulse never cuts a slice off loudly.

spectral — Spectral Clouds+ for the Nebulae

A spectral instrument in the spirit of **WorkshopSpectral** (the Workshop Computer port of Julius Kammerl's **Spectral Clouds**). It is the same idea, grown up for the Nebulae rather than copied.

The patch splits your sound into **log-spaced frequency bands** and gives each band a random gain. A probability threshold shuts the quiet ones, so the sound you hear is your input with **moving holes carved through its spectrum**: few bands give broad, obvious sweeps, and many bands give fine, comb-like shimmer.

New on the Nebulae:

- **true stereo**, with a Spread control;
- **spectral pitch** (1V/oct, ±3 octaves): the frozen spectrum becomes a playable chord;
- **band drift**: the hole pattern scrolls up or down the spectrum like a barber pole;
- **gather**: layer several spectra into one cloud;
- **re-capture**;
- the original's **tape delay**, on the alt layer.

Panel map

```
              [ speed ]                  start   = RE-ROLL RATE  CCW = only on Reset, then every ~3 s .. every frame
 start                          size     size    = BANDS  4 .. 128 (log-spaced)
                                         density = PROBABILITY  how many bands are open
 density       [ pitch ]        overlap  window  = SMOOTHING  CCW = snap, then 50 ms .. 3 s morphs
                                         overlap = TEXTURE  phase scramble: smears into a wash
        blend          window            blend   = dry <-> wet
                                         pitch   = SPECTRAL PITCH, 1V/oct (click = original pitch)
                                         speed   = DRIFT  the band pattern rises (right) / falls (left), centre = still

[record]       [file]           [source]         [reset]          [freeze]
GATHER         RE-CAPTURE       SEALED           RE-ROLL          FREEZE
```

Button / jack	Does	LED
Reset (jack or button)	**Re-roll** every band gain now. Clock it for rhythmic spectral patterns (start fully CCW so re-rolls only come from the clock)	flashes
Freeze	**Freeze** the spectrum: hold the magnitudes (see below). A gate into the Freeze jack toggles it	on = frozen
Source	**Sealed**: frozen or gathered sound keeps its own phases, so nothing of the live input gets through. Hold Source for the alt layer	on = sealed
Record	**Gather**: the spectrum keeps the loudest of what it holds and what comes in, decaying slowly. A gate into the Record jack toggles it	on = gathering
File (button)	**Re-capture**: take the current input as the new held spectrum, without unfreezing	—

All six pots have CV jacks, so every parameter can be modulated. Pitch tracks 1V/oct.

Alt layer (hold Source): start = **stereo spread**; size = **gather decay** (CCW ≈ 1 s, CW ≈ 60 s to silence); window = **tape time** (25 ms – 2 s); overlap = **tape feedback**; density = **tape wow**; blend = **tape mix** (CCW = off).

Alt knobs only take effect once you **move** them, so holding Source never makes the tape or spread jump to where the knob happened to be left by another instrument.

What to feed it

It carves holes in a spectrum, so it needs a **sustained, harmonically rich spectrum** to carve.

- **Great:** pads, drones, held chords, noise (BANDS low + SMOOTHING slow turns noise into a slow chord progression), feedback, cymbals, room tone, whole mixes, distorted guitar, organ.
- **Poor:** drums and sparse plucks (41 ms of latency and a 2048-point window smear transients into a wash), single sine waves (one band does all the work), and anything quiet (the effect scales with the input).

Quick start:

1. Patch noise or a held chord in.
2. Set size about 2 o'clock, window slow (about 2 o'clock), density at noon, blend fully CW.
3. Sweep density and listen to the holes open and close.
4. Turn Speed to its click for a slow rising drift, or patch a clock into Reset.

The band filter

Control	What you hear
density low	sparse, isolated resonant bands: pitched, gated, hollow
density noon	the source clearly present, with moving holes punched through it
density high	nearly the full spectrum, with a subtle shimmer
size low (4–8 bands)	broad, obvious sweeps, close to a random multi-band gate
size high (64–128)	fine comb-like textures
window CCW	gains snap: with a fast re-roll rate this is a stuttering spectral gate
window CW	bands drift open and shut over seconds
overlap	phase scramble: transients smear into a diffuse wash; independent per side, so it also widens the image

- **Silences:** with few bands, a low density can shut every band, so the patch goes quiet through the bottom of density's travel. Turn size up for a smoother density sweep. (The original works the same way.)
- **Density and smoothing:** the threshold follows how spread out the gains currently are, so density opens about the same number of bands whatever smoothing is set to.
- **Low end:** everything below about 190 Hz moves as one band, so bass doesn't chatter at high band counts.

Freeze, Sealed, Gather, Re-capture

Freeze (Source off) holds the **magnitudes** and keeps taking **phase from the live input**. That keeps the hold alive and animated, and what you play into it reshapes it. Like the original, it is an effect on a signal: pull the input and the level drops (measured −9.6 dB) into a thinner sound.

Sealed freeze (Source on) also holds each partial's frequency and keeps rotating it, so the held sound sustains on its own as a chord. Pull the input and it carries on (measured: level holds). This is the one to pitch around.

Gather (Record):

- Unfrozen, the held spectrum keeps the loudest of itself and the input, decaying at the alt-size rate. Sounds hang and blur into each other, like a spectral sustain.
- Frozen, the decay stops: whatever you play **adds** to the hold wherever it's louder. Freeze a chord, turn Gather on, and play another chord into it.
- With Sealed on, gathered partials ring at their own frequencies even after the input stops.

Re-capture (File) replaces the hold with the current input in one frame: a quick "refreeze" without the gap of unfreezing.

The band filter, drift, texture and pitch all keep working on the frozen sound.

Spectral pitch

Pitch moves the whole processed spectrum, 1V/oct, from −36 to +24 semitones (click = no shift):

- Each partial is re-rendered at its new frequency with the right spectral shape, so levels stay put and the result is clean (measured: level within 0.1 dB from −36 to +24, frequency within the analysis resolution).
- The band holes stay where they are in frequency, so the content moves *through* them.

On **sealed frozen** material this turns the hold into a playable chord: sequence it, or sweep it.

On live material it works too, as a spectral pitch shifter, with the usual phase-vocoder softening of transients.

Fast knob sweeps sound a little smeared until the knob settles: phases re-align 40 ms after the last change.

Drift (Speed)

Speed scrolls the band pattern through the spectrum:

- **Centre (0.5):** still.
- **The clicks:** one cycle every 4 s, rising to the right and falling to the left.
- **The ends:** 4 cycles a second, a barber-pole flutter.

Because it is a cycle, the pattern comes back to where it started.

Note: the Nebulae remembers Speed across instruments, so 05 may load already drifting. Turn Speed to the centre for a still pattern.

Stereo

Left and right are analysed separately, so the input's stereo image is kept.

Alt start	Pattern
CCW	both sides share one band pattern
CW	the right side has its own pattern, for wide, unlinked holes
in between	blends the two

Texture also decorrelates the sides.

Tape delay (alt layer)

This is the original's TAPE mode, after the spectral engine: the holes are carved first, then the delay carries the result off. It is deliberately tape-like:

- two-LFO wow and flutter;
- each repeat gets darker (damping in the loop);
- a loop highpass stops the low end piling up;
- the feedback path soft-clips;
- feedback is capped at 0.85, so tails resolve rather than freeze. (The patch already has freezes for that.)
-

Moving the time while repeats are sounding gives the tape-speed swoop.

Differences from WorkshopSpectral

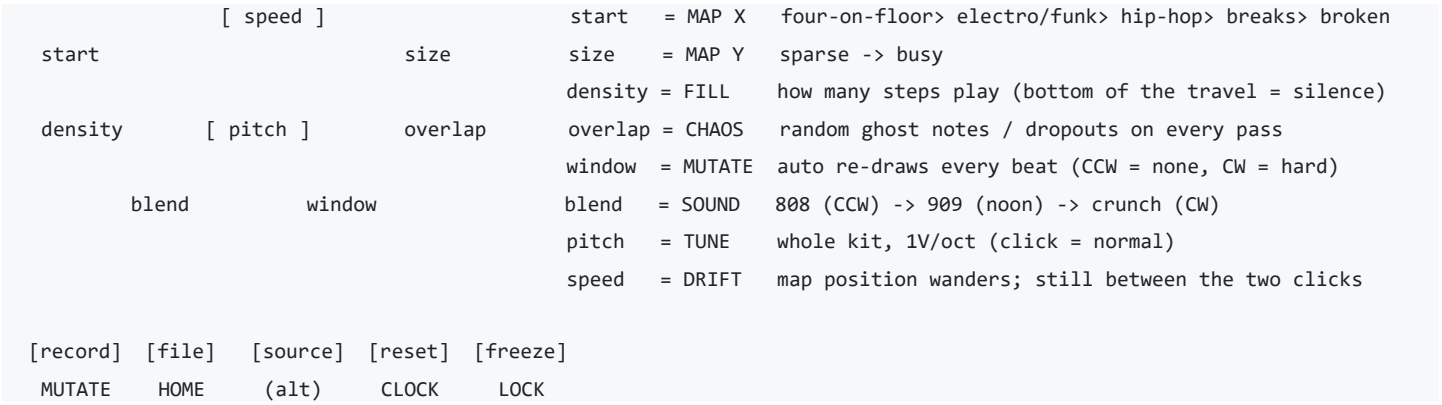
	WorkshopSpectral (Workshop Computer)	**05_spectral (Nebulae)**
FFT	1024 points, mono out	2048 points (finer, cleaner low end), true stereo
Latency	21 ms	41 ms (the dry path is delayed to match, so blend never combs)
Sealed freeze	hold the switch Down 2 s	Source latch (the LED shows it)
Re-roll	Pulse In 1 + X knob	Reset jack/button + start
Output level	Y on page 2	soft clip on the output (fixed level). Use blend and your mixer
Pitch, drift, gather, re-capture, spread	—	new
Tape	alt-boot mode, replaces page 2	always available on the alt layer
CV / pulse outs (density, re-roll pulse, load)	yes	not possible: Pd can't drive the Nebulae's outputs or LEDs

topo — Topo drums for the Nebulae (user guide)

A clocked **drum machine that writes its own patterns** and plays them on its **own analog-style drum synth**. Send it a clock and it plays. The patterns come from a map of drum styles that you move through with two knobs (the idea of Mutable Instruments' Grids, with our own map). They can **mutate**, by hand or on their own, and you can always go back **home**.

- **Six voices:** kick, snare, clap, closed hat, open hat (choked by the closed hat), perc/tom. All are synthesised, so there are no samples to load.
- **The map:** 25 hand-written patterns on a 5 × 5 grid, from four-on-the-floor to broken polyrhythms, and from sparse to busy. Map X/Y blend smoothly between neighbouring patterns.
- **Mutation:** Record re-draws a few steps; Window re-draws some every cycle; Speed makes the map position drift. Freeze locks it all.
- **Sound:** one knob morphs between three hand-designed kits: 808-ish, 909-ish, crunchy. The sounds never change unless you turn it.

Panel map



Button / jack	Does	LED
Reset jack	**Clock in: one pulse = one 16th step.** The button is a manual step (tap through the pattern)	flashes
Source gate jack	**Back to step 1:** the next clock plays the pattern's first step (send it once per bar or phrase)	—
Freeze (button or gate)	**Lock:** no automatic mutation, no drift. Record still mutates	on = locked
Record (button or gate)	**Mutate now:** re-draws a burst of steps (4, up to 16 with Window turned up). Every gate edge mutates once. The LED just toggles; ignore it	—
File (button)	**Short press: go home.** Brings back the home pattern. **Hold 1 s: store** the current pattern as home. **Double-tap: vanilla**, the plain map with no mutations and no drift (your home is kept). Home at load = the plain map	—
Source button	Hold for the alt layer. Its latch does nothing here (the Source gate jack also toggles it)	—

All six pots have CV jacks. Sweeping **Map X/Y** with an LFO or envelope is the classic way to play this kind of sequencer. **Fill** CV makes builds and breakdowns. **Pitch** tracks 1V/oct for the whole kit.

Alt layer (hold Source):

Knob	Does	Default
start	**Kick fill** offset (± around the main Fill)	centre = none
size	**Snare + clap fill** offset	centre
density	**Hats fill** offset	centre
overlap	**Perc fill** offset	centre
window	**Length** 4 / 8 / 16 / 32 steps (four zones; always 4/4)	32 (top quarter)
blend	**Decay**: the whole kit ×0.25 (tight) – ×4 (boomy)	×1 at centre
speed encoder	**Swing**: click = straight; 8 detents either way = full **12/8** (triplet); in between = in between	straight

Alt knobs (and alt Speed) only act once you **move** them, so holding Source doesn't make anything jump to where another instrument left them.

Swing and input gain: alt Speed is also the Nebulae's **input gain**. 06 has no audio input, so that doesn't matter here, but the gain carries over to the next instrument. **Click alt Speed (straight = 0 dB) before switching instruments**, or swing CCW: that only ever lowers the input gain, while CW raises it.

First steps

0. Clean slate.

- **Freeze** LED off (tap it if it's lit).
- **blend** fully CCW (the 808 kit). The Nebulae remembers blend from the last instrument, which usually leaves it fully CW (the crunch kit).
- **Speed** between its two clicks (no drift). It snaps to the clicks.
- **Pitch** on its click.

1. A beat.

1. Patch a clock into **Reset**, one pulse per 16th note. For 120 BPM that is 8 pulses per second (a ×4 clock from a quarter-note clock).
2. Set **start** fully CCW, **size** at noon, **density** at noon, and **overlap**, **window** and **blend** fully CCW.
3. You hear a four-on-the-floor house beat: kick on every beat, clap + snare on 2 and 4, open hats on the off-beats. (No clock? Tap **Reset** to step through it by hand.)
4. Patch a once-per-bar trigger into the **Source gate** jack, so step 1 lands on your downbeat.

2. Fill. Turn **density** down: the beat strips to its skeleton (kick, backbeat), then to silence. Turn it up: ghost notes, 16th hats and perc fills appear, in order of importance. Every pattern is written so the lowest fill is the core of the groove.

3. The map. Turn **start** (X) slowly from CCW to CW: house → electro/funk → hip-hop → breakbeat/jungle → broken polyrhythms. Turn **size** (Y): sparse at CCW, busy at CW. Positions between the 25 written patterns blend their neighbours, so every knob position is a groove.

4. Chaos. Bring **overlap** up a little: each pass adds or drops random ghost notes. Nothing is stored, so turn it back down and the pattern is clean again.

5. Mutate by hand. Press **Record**: a few steps change, and they stay changed. Press it again for more. Too far? A **short File press** brings home back, and a **double-tap** goes all the way back to the vanilla map.

6. Find a home. When a mutated pattern is good, **hold File for a second**: that is now home. Keep mutating; a short press returns to it.

7. Let it evolve, or mutate it hard. Turn **window** up and the pattern re-draws steps on every beat:

- low settings: an occasional change, a slow evolution;
- noon: about one change per beat;
- fully CW: up to eight per beat, so about two thirds of the hits change from one 2-bar cycle to the next.

It keeps roughly the density Fill sets, so it scrambles the groove rather than just filling it in. Press **Freeze** to hold what you have (press again to carry on). Double-tap **File** for vanilla.

8. Drift. Turn **Speed** past either click: the map position wanders by itself around where X/Y are set, drifting between styles. The further from centre, the faster. Freeze stops it.

9. Sound. Turn **blend**:

- **CCW, 808-ish:** a long boomy kick, a tonal snare, and the 808's six-oscillator metal hats.
- **Noon, 909-ish:** a punchier, slightly driven kick, a snappier snare, brighter hats.
- **CW, crunch:** a short, hard-driven kick, a noisy snare, high hats, a rim-like perc. Past noon a **crunch bus** fades in on the whole kit: drive into a hard clip, a bit-crush down to 6 bits, and sample-rate reduction down to 6 kHz, all getting stronger towards CW. Its level is compensated, so turning into crunch doesn't make the kit louder.

Positions in between blend the two neighbouring kits, and any position is repeatable. The sounds don't change unless the knob (or its CV) moves. Pitch tunes the whole kit, and alt **blend** sets the decay.

How the patterns work

Each of the 25 map patterns gives every step of four lanes (kick, snare, hats, perc) an **importance** level. **Fill** sets the level a step needs to play, so low Fill keeps only the most important hits. The strongest snare steps also fire the **clap**, and the strongest hat steps are **open** hats (a closed hat chokes an open one).

Mutations are stored as changes on top of the map, so X/Y still move the groove while your mutations come along. Each mutation picks a random lane and step, then does one of three things:

- **Move** (half of them): swap a playing step with a silent neighbour up to 2 steps away. The hit moves in time.
- **Flip** (30%): turn a playing step off, or a silent step on. Turning on only happens some of the time (more at high Fill), so heavy mutation keeps about the density Fill sets.
- **Heal** (20%): remove the change on that step, so heavy mutation doesn't turn into noise.
-

Home stores those changes and the drift position. **Length** shortens the loop to 16, 8 or 4 steps, so it stays in 4/4. **Odd meters come only from a clocked reset:** send a trigger to the **Source gate** jack every N steps. For example every 12 steps for 3/4 or 12/8, or every 20 for 5/4. Stop the trigger and the loop goes back to the full Length.

Differences from Grids

- **Our own map**, not Grids' (Grids' code is GPL; this repo is MIT). It is laid out by style on X and density on Y, with four lanes instead of three, and a 25-node grid like Grids'.
- **Built-in drum synth** with six voices, instead of trigger outputs (the Nebulae has no gate outs for Pd).
- **Mutation, home, lock and drift** are new. Chaos works like Grids' randomness.
- **Clock = 16ths only** (Grids also takes 24 ppqn). Swing and Length are on the alt layer.
- No accent outputs: accent is the velocity of each hit (stronger steps play louder).

shimmer — shimmer reverb

A Strymon-style **shimmer reverb**. A lush 8-line reverb has a pitch shifter in its feedback, so every pass climbs another interval and the tail blooms into an octave-up halo. Two pitch voices are available, and there's an option for the SoS-style "pre" shimmer: one layer, no climb. There's also tank modulation into ensemble chorus, freeze, swell, ducking and clear.

Panel

Control	Function
Mix (Blend)	Dry/wet, equal-power. Fully CW = 100 % wet (use this when it sits on a mixer send)
Size	Decay time: 0.4 s → 60 s (noon ≈ 5 s)
Window	Tone: high-frequency damping in the reverb. CCW = dark, fully CW = no damping
Start	Pre-delay 0 → 1 s (squared curve: noon = 250 ms)
Density	**Shimmer amount**. 0 = plain reverb. Noon = classic halo. CW = dense, fast-climbing shimmer
Overlap	**Mod**. CCW → noon: slow modulation inside the reverb (lush, no metallic ring). Noon → CW: also fades in a 3-voice ensemble chorus (Dimension-D / string-machine feel)
Pitch (+ V/oct)	**Voice A interval** in semitones. **The click = +12 (octave up)**. Turn to change the interval (−24 … +36), and use V/oct to sequence it
Speed	**Voice B interval**. Between the two clicks = **off**. Past the right click: +5, +7, +12, +19, +24. Past the left click: −5, −12, −24 (sub-octaves). Speed carries over from the last instrument, so check it after loading

Button / jack	Function
Freeze (LED)	Infinite hold. New input is shut out, the tail sustains forever, and the shimmer pauses so the held sound doesn't drift away
File (hold)	Momentary freeze while held
Source (LED)	Off = **Loop** shimmer (cascading, the Strymon sound). On = **Pre** shimmer: the input is pitch-shifted once before the reverb, giving one octave-up layer and no climb
Record (LED)	**Swell**: each new note fades into the reverb over about 1 s, for bowed, volume-pedal entries. A note counts as new after about 150 ms of silence
Reset (button or gate jack)	**Clear**: the tail is cut and the reverb is empty about 0.4 s later. From a gate it chops the tails rhythmically

Alt layer (hold Source)

An alt knob takes effect only once you move it.

Control	Function	Default
alt **Start**	Low cut, 20 Hz → 1 kHz, on the reverb input and the shimmer loop	80 Hz
alt **Size**	Room size, 0.5× (small, plate-like) → 2× (cathedral). Changing it bends the pitch briefly, like tape	1× (noon)
alt **Density**	Shimmer tone: a lowpass on the shifted signal, 1 → 16 kHz. Lower = smoother and darker, CW = glassy	6 kHz
alt **Overlap**	Mod / chorus rate, 0.05 → 5 Hz. Depth backs off automatically above 1 Hz	0.4 Hz
alt **Window**	Voice A / B balance: CCW = A only, noon = equal, CW = B only. This only matters when Voice B is on	noon
alt **Blend**	Ducking: the wet signal dips under your playing (up to −24 dB) and blooms back when you stop	off
alt **Pitch**	Detune: spreads the shimmer voices up to ±25 cents (L/R and A/B), which makes it cloudier and wider. The alt-reset click sets it back to 0	0
alt **Speed**	Not used (it's the Nebulae's input gain)	—

RESET ALT LAYER: Hold **Source** and press **Record**: every alt setting below goes back to its default. The knobs then do nothing until you move them again.

Detune follows the Pitch encoder's own value as soon as you turn it again, so also click Pitch while holding Source (the Nebulae's own alt reset) to zero it on the module side.

Quick start

1. Mix about 2 o'clock, Size noon, Window 2 o'clock, Density at 0, Pitch on its click (+12), Speed between the clicks.
2. Play. It's a plain hall. Bring **Density** up towards noon and the octave-up halo blooms in behind each note.
3. Push Size past 3 o'clock and Density higher for the endless rising pad. Pull alt Density (shimmer tone) down if it gets glassy.
4. Turn Speed past the left click to the first step past −12 for octave-up plus sub-octave (organ-like). The first step past the right click gives +5 (a fourth); the next step gives +7, the fifth.
5. **Freeze** to hold a chord and play over it. Reset clears it.

Notes

- The shimmer decays with the reverb. At high Density the tail is a little shorter than the Size setting, because some energy climbs out of range through the shimmer lowpass.
- The loudest settings (Size and Density fully up, sustained input) are bounded by soft clipping. They're loud but safe, with no runaway.

resonator — exciter → resonator → delay

A four-voice **resonator synth/effect** after the Morphor ECHON. An **exciter** (a VCO shape, noise or crackle, shaped by an envelope, or your audio input) is fed into a **pitched delay-line resonator** that rings like a string, a tube or a metal bar. The result then goes through a dark, wobbly **BBD-style stereo delay**. Strike it with Reset, play it with V/oct, or run any audio through it as a tuned resonator. The output is stereo. The audio inputs are only used when Source is on.

Panel

Control	Function
Pitch (+ V/oct)	Note, continuous 1V/oct. **The click = C4** (261.6 Hz). The knob spans C1 … C6, and CV goes further, up to 2.6 kHz (1.3 kHz inverted). What it moves depends on the **V/oct mode** (File, below)
Start	Exciter material. CCW → CW: sine → triangle → saw → square → noise → sample-and-hold crackle, crossfading between neighbours
Size	Decay: 0.05 s → 30 s (noon ≈ 1.2 s), the same at every pitch
Density	Exciter envelope. CCW = a short pluck/strike. Noon = a softer hit (≈ 30 ms attack). CW = a slow bowed swell (1.5 s in, 3 s out)
Window	Damping inside the resonator, which follows the note. CCW = muted, woody, short; noon = a natural string; CW = bright and metallic, ringing for the full Size
Overlap	Delay feedback. Past about 3 o'clock it runs away into BBD-style self-oscillation, which soft clipping keeps in check
Blend	Delay mix (equal-power). CCW = no delay; CW = delay only
Speed	Delay time, 20 ms → 2 s. Turning it while echoes play bends their pitch, like a BBD. Speed carries over from the last instrument, so check it after loading

Button / jack	Function
Reset (button or gate jack)	**Strike**: fires the exciter envelope. Patch a trigger or clock here. Strikes wait 3 ms so a V/oct change sent with the gate lands first
Freeze (LED)	**Hold**: the resonator rings forever at its current tone, and new strikes are shut out. Pitch still bends the held note
Source (LED)	**Audio in**: your input becomes the exciter instead of the internal one. Drums, voice and synths ring at the resonator's pitch
Record (LED)	**Invert** the feedback. The tone becomes hollow and clarinet-like (odd harmonics only), and it stays on the same note
File short press	**V/oct mode**: Coupled → Res only → Exc only → Coupled. See below. It starts on Coupled at load. There's no LED for it, so count your presses
File held (≥ 0.3 s)	**Delay hold** while held: the echoes repeat forever and the delay input is shut

V/oct mode (File short press)

Mode	Resonator	Exciter	Feels like
Coupled (default)	follows Pitch	follows Pitch + the offset	the ECHON: both play the note
Res only	follows Pitch	fixed at C4 + offset	a fixed drone, noise or crackle whose overtones a moving string picks out
Exc only	fixed at C4 + offset	follows Pitch	a fixed tuned body that you play notes through: they bloom near its pitch and its harmonics

The **offset** is alt Window: semitones, ±36, noon = 0. In Coupled it's the exciter-to-resonator interval (e.g. +12 or +7 for brighter, bell-like strikes). In the other modes it's the fixed side's note.

Alt layer (hold Source)

An alt knob takes effect only once you move it.

Lost? Hold Source and press Record: every alt setting below goes back to its default, and the V/oct mode goes back to Coupled. The knobs then do nothing until you move them again.

Control	Function	Default
alt **Start**	Exciter lowpass (12 dB), 150 Hz → open. Softens the strike: felt hammer instead of pick	open
alt **Size**	Stereo spread: detunes L/R by up to 30 cents (chorus-like beating)	6 cents
alt **Density**	Auto-repeat: CCW = off, then 0.25 → 16 Hz self-strikes (the ECHON's looping envelope). Fast rates with a long Density bow continuously	off
alt **Overlap**	Delay tone: CCW = murky BBD (1.5 kHz, more wobble), CW = clean (16 kHz, no wobble)	≈ 10 o'clock
alt **Window**	**Offset**, ±36 semitones in steps, with a wide 0 around noon (see V/oct mode)	0
alt **Blend**	Exciter dry ↔ resonator. CW = resonator only. Turn it back to hear the raw strike or input as well	CW
alt **Pitch**	Delay ping-pong: 0 = straight stereo, up = echoes bounce L ↔ R. The Nebulae's alt-reset click sets it to 0	0
alt **Speed**	Not used (it's the Nebulae's input gain, which matters with Source on)	—

Quick start

1. Pitch on its click, Start at 10 o'clock (triangle), Size noon, Density CCW, Window noon, Blend CCW.
2. Press **Reset**: a plucked string. Sweep **Window** (muted → bright) and **Start** (hammered → noisy breath).
3. Density CW and press Reset again: a bowed swell. Turn on auto-repeat (alt Density) for an endless bow.
4. Patch a sequencer: CV → Pitch (V/oct), gate → Reset. Bring in **Blend** and **Overlap** for the delay, and set **Speed** to taste.
5. Try **Record** (invert) for hollow, clarinet-ish tones, and **Freeze** to hold a note while you bend Pitch.
6. **Source** on and feed it drums: every hit rings at the Pitch note. Press File once (Res only) and sequence the ringing pitch while the drums play on.

Notes

- Pitch is accurate: it measured within 0.01 cents offline from 92 Hz to 2.5 kHz, and within about 1 cent with Window fully CCW. The damping is compensated, so turning Window doesn't detune.
- With Window CCW, high notes are short whatever Size says. Each trip round the loop loses energy in the damping filter, and high notes make more trips per second, like a muted string. Size is exact with Window CW. At noon it's exact except for long Sizes on high notes, which come out a little shorter.
- Volume: loud, sustained resonances (bowed, auto-repeat, matched audio input) are held near −9 dB RMS by an automatic level on the excitation. Single strikes aren't affected. The worst cases are bounded by soft clipping.
- **4-note polyphony with oldest-note stealing, and a ringing note keeps its pitch when V/oct moves. Owner's picks: mono voices panned (alt Size = width), an exciter per voice, audio input into all 4 voices, pitch latched at the strike + a 10 ms catch.**

Differences from the ECHON

- Four voice, not six. A new strike re-excites the same string, which is still ringing.
- There's no 12 dB resonator input filter as a fixed knob. alt Start does that job, and Window is the 6 dB loop filter.
- No LFO/mod matrix. Every knob has its own CV jack instead.
- There's an echo after the resonator (the ECHON has none), plus the V/oct routing modes.

SuperCollider instruments for the Nebulae (uf2)

These run in the **SC bank**: hold Speed, turn Speed clockwise **past Pd**, so the right Speed light turns **blue**. Then press a lit button. Expect about 5–10 s of silence while SuperCollider starts.

File	What it is	Needs
`01_sctest`	Test patch: a plucked saw for checking the controls	Reset to play
`02_sc_drift`	Evolving drone / pad	Nothing, it sounds at once
`03_sc_strike`	Plucked string with a resonant body	Reset (button or gate), or Record for auto-pluck
`04_sc_cloud`	Granular clouds from the audio input	Audio patched in
`31-simple`	Technobear's example: a saw drone. **Loud: turn the output down first**	Nothing

Each control's CV jack adds to its knob, as on every Nebulae instrument. Pitch follows 1 V/oct, and a Pitch click returns it to the centre.

02_sc_drift: drone / pad

Five detuned voices, a morphing oscillator and a drifting resonant filter.

Control	Does
Pitch + V/oct	Pitch, 1 V/oct (click = C4)
Start	Oscillator: saw → pulse → sine
Size	Detune spread: unison → wide
Density	Filter cutoff (80 Hz – 12 kHz)
Window	Filter resonance
Overlap	Drift depth: the filter and pulse width move
Speed (encoder)	Drift rate (slow ↔ fast)
Blend	Level
Freeze	Sub-octave on/off
Source	Chord: adds a fifth and an octave
Reset	Swell: the sound dips and blooms back

Try: Start fully clockwise (sine), Density low, Freeze on for a deep sine drone. Start at noon (pulse), Window up, Overlap up and Speed slow for a moving resonant pad.

03_sc_strike: plucked string

Karplus-Strong string with a pluck-position comb and a wooden body.

Control	Does
Reset (button or gate)	Pluck. A clock or trigger sequence into Reset plays it
Pitch + V/oct	Pitch, 1 V/oct (click = C4). A sequencer into Pitch CV + a gate into Reset = a melody
Size	Decay (0.3 – 20 s)
Density	Brightness of the pluck
Window	Body: dry string → resonant wooden body
Start	Pluck position: thin (near the bridge) → round (middle)
Overlap	Second string: detune and stereo width
Record	Auto-pluck on/off (internal clock)
Speed (encoder)	Auto-pluck rate (0.5 – 16 per second)
Freeze	Sustain: the string rings on
Source	Chord: adds a fifth
Blend	Level

04_sc_cloud: granular input

A 4-second buffer records the input all the time, and grains read it back pitched and scattered.

Control	Does
Audio in	The source (stereo summed to mono)
Start	How far back the grains read (just recorded → 4 s ago)
Size	Grain length (20 – 500 ms)
Density	Grains per second (1 – 80). The last tenth gives random (dense) grains instead of regular
Window	Position scatter around Start
Pitch + V/oct	Grain pitch, 1 V/oct (click = original pitch)
Overlap	Stereo spread of the grains
Speed (encoder)	Pitch scatter: centre (1×) = none; either way = more random detune
Blend	Dry / wet
Freeze	Stop recording: the buffer holds what it has
Reset	A burst of grains

Control	Does
Source	Reverse: grains play backwards

Try: patch a melody in, Blend fully wet, Density at noon, Size small, then Freeze and play with Start and Pitch.

pearls — Mutable Instruments Beads on the Nebulae

Pearls is Mutable Instruments **Beads**, ported to the Nebulae by XponentOneAudio (MIT), with the original Beads DSP and a few extras. It is a granular processor: it records the input into a buffer all the time and plays **grains** (short snippets) from it. Density decides how many and how regularly, Time how far back they read, Size how long they are, Shape their envelope, and Pitch their pitch. Fully clockwise, Size turns it into a delay. Feedback, dry/wet and reverb come after the grains. Freeze stops the recording. With no input, Source turns it into a wavetable synth.

It is in the **Csound user bank** as `pearls`. It needs `pearls.instr` and the `csound_plugins` folder (with `libpearls.so`) on the stick. The first load after installing uf2 installs the plugin and takes a few seconds longer. Features marked *new* came in Pearls 5–8 and haven't been tested on hardware by their author yet.

Knobs and encoders

Each CV jack adds to its knob.

Nebulae	Pearls (Beads)
Start	TIME: how far back in the buffer the grains read
Size	SIZE: grain length. Left of about 11 o'clock, grains play backwards. **Fully clockwise = delay mode**
Window	SHAPE: grain envelope, from sharp to smooth
Density	DENSITY. **12 o'clock = no grains.** Clockwise = more, at random times. Anticlockwise = more, regularly
Pitch + 1 V/oct	PITCH, ±48 semitones. Click = unison
Overlap	FEEDBACK
Blend	DRY/WET
Speed	REVERB amount: turn clockwise from 0. Click toggles a light reverb on/off

Buttons and gates

Nebulae	Pearls
Freeze button / gate	FREEZE: stops recording, the grains keep playing the buffer (the gate holds it while high)
Reset button	SEED: a tap plays one grain (in delay mode, tap tempo). **Re-lock** the grain generator: hold 4 s and release, or press Freeze while holding
Reset gate	SEED input: each trigger plays a grain; a clock plays grains in time (counts as patched while edges arrive, 4 s timeout)
File button / gate	QUALITY, on release: hi-fi → cold digital → sunny tape → scorched cassette → hi-fi. **Hold 1.5 s**: restart the automatic input gain
Source button / gate	Wavetable synth mode on/off
Record button	TAP TEMPO *new*: tap 2+ times to clock the grains; later taps re-align; hold 1 s to stop
File held + Record	PITCH SCALE *new*: off → chromatic → major → minor → major pent. → minor pent. → octaves + fifths → octaves → off
EOL out	A gate per grain (in delay mode, per delay tap)

LEDs

LED	Shows
Reset	Grain activity (dim while locked and idle)
Freeze	Frozen (button, gate, or after loading a sample)
File	Lit while pressed. **Slow blink**: the automatic gain is measuring the input (~5 s after start or a new input). Fast blink: loading or saving
Record	Tap-tempo beat
Encoder LEDs, after a quality change	hi-fi = white, cold digital = cyan-green, sunny tape = amber, scorched cassette = purple
Pitch LEDs, after a scale change	off = white, chromatic = red, major = amber, minor = yellow, maj. pent. = green, min. pent. = cyan, oct + 5th = blue, octaves = magenta
All encoder LEDs after load/save	green = loaded, blue = saved, red = failed

Secondary layer (hold Source)

Nebulae	Pearls
Start / Size / Window	Randomness of TIME / SIZE / SHAPE (fully anticlockwise = none)
Pitch encoder	Randomness of PITCH
Density	Input gain: fully anticlockwise = automatic (default), else 1×–40× manual
Overlap	Reversed grains *new*: more reversed towards clockwise
Blend	Stereo spread *new*
Speed encoder	The Nebulae input level (as usual)
Freeze	Randomness shape: off = even, on = mostly small with occasional big jumps
Record	Mono buffer: left input only, twice the time
File	**Load** the next WAV from the stick into the buffer *new* (it loads frozen)
Reset	**Save** the buffer to the stick as `pearls_001.wav`, … *new*

Changing quality, mono or synth mode **clears the buffer**, as on Beads. While frozen, those changes wait until you unfreeze, so a loaded sample is safe. Recording time runs from about 4 s (hi-fi stereo) to about 32 s (scorched cassette, mono). The quality and a frozen buffer survive a power cycle.

Quick start

1. Patch audio in. Blend at noon. Watch the File LED blink slowly for a few seconds (gain measuring).
2. Density a little clockwise: random grains. A little anticlockwise: a steady stream.
3. Turn Start (how far back), Size (short clicks to long smears; left of 11 o'clock = reversed) and Window.
4. Freeze, then play with Pitch and Start on the frozen buffer.
5. Size fully clockwise for the delay; Reset taps set its time.
6. Tap File through the four qualities; hold Source + File to load a WAV from the stick.

chop — slice sequencer

A Pure Data instrument that cuts a loop into slices at its hits and plays them back in a new order, in time with your clock. The loop is either something you grab from the inputs (Record grabs exactly 1, 2 or 4 bars, starting on a bar line) or a WAV from the stick (File). Turned all the way down, chop plays the loop back exactly as it was. Turning the knobs up shuffles the hits, drops some, repeats them as ratchets, plays them backwards and makes them jump in pitch. The rhythm of the loop always stays: only what plays on each hit changes.

It's in the **Pd bank** as `K_chop` (yellow page, Record button), file `09_chop.pd` in the repo.

Panel map

```
          [ speed ]            speed   = PITCH JUMPS  off between
start                  size               the clicks, R up, L down
                               start   = SHUFFLE   own -> any slice
density    [ pitch ]  overlap  size    = GATE      stabs -> full
                               density = HITS      none -> all
      blend     window         overlap = RATCHET   chance of 2-6x
                               window  = REVERSE   chance
                               blend   = input <-> chop
                               pitch   = TRANSPOSE, 1V/oct

[record]  [file]    [source]  [reset]   [freeze]
 GRAB      WAV       STEP 1    CLOCK     HOLD DICE
```

Controls

Control	Does
Start	Shuffle: the chance that a hit plays another slice of the loop instead of its own. Fully anticlockwise = the original loop; fully clockwise = every hit picks any slice
Size	Gate: how much of each slice plays. Fully clockwise = the whole slice (a perfect replay); anticlockwise = short stabs (down to 12 ms)
Density	Hits: the chance that a hit plays at all. Fully clockwise = every hit; turning down drops hits out (silence until the next one)
Overlap	Ratchet: the chance that a hit repeats 2, 3, 4 or 6 times within its 16th (the counts are set on alt Overlap). The last repeat plays on
Window	Reverse: the chance that a hit plays backwards (from the end of its slice)
Blend	Input ↔ chop (equal power). Fully clockwise = only chop
Pitch + 1 V/oct	Transpose every hit, in semitones; the click = original. Like a sampler: higher is also shorter
Speed	Pitch jumps. Between the two clicks: none (a load on either click is safe). Clockwise of the right-hand click: a growing chance that a hit jumps **up** (+5, +7 or +12). Anticlockwise of the left-hand click: **down** (−12 or −24, a tape drop)

Button / jack	Does	LED
Reset (button or gate)	Clock in: **one pulse = one beat** (a quarter note). chop runs 16th steps inside the beat. Tap the button twice for a tempo by hand. With no clock it runs at 120 BPM from load, and keeps the last tempo when the clock stops. If the clock stops for over 2 s, the next pulse is step 1 (so starting your sequencer starts chop's bar)	—

Button / jack	Does	LED
Record (button or gate)	Grab: the next 1, 2 or 4 bars of the input (alt Start), starting at the next bar line. When the grab ends, at a bar line, it plays at once. A press during a grab is ignored	toggles (no meaning)
File	Short press: the next `.wav` / `.aif` on the stick (FLAC can't be read by Pd). It plays from the next bar line. **Held ½ s**: back to the last grab	—
Source (button or gate)	Step 1: the next beat becomes step 1. A gate up to 30 ms after a beat re-plays that beat as step 1, so a gate sent together with the clock is never a beat late. Holding Source still opens the alt layer	toggles (no meaning)
Freeze (button or gate)	Hold the dice. Off: a new pattern is rolled at the start of every cycle. On: the same pattern repeats. The knobs still work while it's held: turning Shuffle up adds swaps (the same ones each time), and turning it back takes them away again	on = held

Alt layer (hold Source)

Control	Does	Default
alt **Start**	Grab length: 1 / 2 / 4 bars (three zones)	1 bar
alt **Size**	Slicing. Fully anticlockwise = equal 16ths. Then onset detection, more sensitive clockwise (more slices). It re-slices at once	noon
alt **Density**	Swing on the 16ths: straight → heavy (the off-16ths up to half a 16th late)	straight
alt **Overlap**	Ratchet counts: anticlockwise 2× only → 2–3× → 2–4× → clockwise 2, 3, 4 or 6×	2–4×
alt **Window**	Slice edges: tight 1 ms fades → soft 20 ms fades	1.5 ms
alt **Blend**	Stutter: the chance that a hit repeats the slice of the hit before it (beat-repeat)	off
alt **Pitch**, alt **Speed**	Not used (alt Speed is the Nebulae's input gain)	—
Source + Record	All of the above back to their defaults	—

Alt values only count once you move the knob (holding Source shows the old values first).

Quick start

1. Patch a clock (one pulse per beat) into **Reset** and a drum loop or drum machine into the inputs. Load `K_chop`. Set Blend fully clockwise, Size fully clockwise, Density fully clockwise, the rest anticlockwise.
2. Press **Record** just before a bar starts. One bar later chop plays the grab back: it sounds like the input.
3. Turn **Start** up: the hits get swapped around. Every cycle brings a new pattern. Like one? Press **Freeze**.
4. Add **Overlap** for rolls, **Window** for reversed hits, **Speed** past a click for pitch jumps, and pull **Density** and **Size** down for space.
5. Or press **File** to chop a loop from the stick instead (no clock needed: it runs at 120 BPM until one comes).

How it slices

- When a grab ends or a file loads, chop looks for **onsets** (sudden rises in level). Each 16th step that has an onset near it starts a slice, and the slice starts 1.5 ms before the onset, so no attack is cut. Steps without an onset let the previous slice play on: this is how the loop's rhythm stays when hits are swapped.
- If it finds fewer than 2 onsets per bar (a pad, a drone), it uses equal 16ths instead. So does alt Size fully anticlockwise.
- A WAV doesn't say its tempo. chop takes its length against the current tempo as 1, 2 or 4 bars (the nearest), and plays each slice at its own speed, starting on its step. So a file follows the clock without changing pitch, like a sliced break. Files longer than 6 bars give their first 4 bars.
- Files at any sample rate play at the right pitch; mono files come out of both sides.

Notes

- A hit sounds about 1.5 ms after its step: that's the pre-roll before the onset. There's no other added latency, and the dry input is not delayed.
- Loading a file reads it in one go, so expect a very short dropout when you press File (Pd can't read files in the background). Grabs have no dropout.
- Up to 43 s per grab or file. chop keeps two grabs and two files in memory, so what's playing is never overwritten by what's loading.
- CPU, measured offline: about the same as glitch.

vocoder — 20-band channel vocoder

A Pure Data instrument that puts the spectrum of one sound onto another, the classic robot-voice and talking-synth effect. The **left input** is the modulator: a voice, drums, anything with a shape. The carrier is either an internal synth chord (a supersaw on V/oct) or whatever you patch into the **right input**. Freeze holds a vowel so you can play it as a chord, Speed shifts the formants up or down, and Track makes the internal chord follow the pitch of the voice.

It's in the **Pd bank** as `L_vocoder` (yellow page, File button), file `10_vocoder.pd` in the repo.

Panel map

```
           [ speed ]            speed   = FORMANT SHIFT  none between
start                  size               the clicks, R up, L down
                                start   = CHORD   8 shapes
density    [ pitch ]  overlap   size    = BANDS   smooth -> sharp
                                density = DETUNE  supersaw width
      blend     window          overlap = SIBILANCE  noise for s/t
                                window  = ENVELOPE   snappy -> slow
                                blend   = voice <-> vocoder
                                pitch   = CHORD ROOT, 1V/oct

[record]  [file]    [source]  [reset]   [freeze]
 TRACK     HOLD      EXT CAR   NEXT CHD  FREEZE
```

Controls

Control	Does
Start	Chord shape of the internal carrier, in 8 zones: unison, octaves, fifth, major, minor, sus4, maj7, min7
Size	Band sharpness: anticlockwise = smooth, wide bands; clockwise = narrow, ringing, more "robot"
Density	Detune: three saws per chord note, spread from none (clean) to a wide supersaw
Overlap	Sibilance: adds noise wherever the voice has "s" and "t" sounds (above 5 kHz), so consonants stay clear
Window	Envelope speed: anticlockwise = snappy (follows every syllable); clockwise = slow, smeared
Blend	Dry modulator (left input) ↔ vocoder (equal power)
Pitch + 1 V/oct	Root of the internal chord. The click = C3. With Track on, it transposes the tracked pitch in semitones instead
Speed	Formant shift. Between the two clicks: none (a load on a click is safe). Clockwise: formants up, to 6 bands (two octaves) at the end, the chipmunk side. Anticlockwise: down, the giant side

Button / jack	Does	LED
Freeze (button or gate)	Hold the spectral envelope: the vowel or sound at that moment stays, and the carrier keeps playing through it, so you can change chords and notes on a frozen vowel	on = frozen
File (held)	Momentary freeze, for as long as you hold it	—
Reset (button or gate)	Next chord shape (one step past what Start says; eight steps go round). A sequencer gate steps through the chords	—
Record (button or gate)	Track: the internal chord's root follows the pitch of the voice on the left input, rounded to the nearest semitone (a robot choir). Pitch then	on = tracking

Button / jack	Does	LED
	transposes it	
Source (button or gate)	Carrier: off = the internal chord; on = the right input (any synth, pad or noise you patch in)	on = external

Alt layer (hold Source)

Control	Does	Default
alt **Start**	Stereo width: even bands to the left, odd bands to the right	half
alt **Size**	Presence: lifts the voice's top end before the bands, for clearer words	a little
alt **Density**	Whisper: blends the internal saws into noise	off
alt **Overlap**	Carrier leak: some of the unvocoded carrier underneath	off
alt **Window**	Track glide: how smoothly the tracked chord moves (0–500 ms)	30 ms
alt **Blend**	Noise gate threshold (−80 … −20 dB): below it the bands close, so hiss doesn't play the carrier	−60 dB
alt **Pitch**, alt **Speed**	Not used (alt Speed is the Nebulae's input gain)	—
Source + Record	All of the above back to their defaults	—

Alt values only count once you move the knob (holding Source shows the old values first).

Quick start

1. Patch a voice (or a drum loop) into the **left** input. Load `L_vocoder`. Blend fully clockwise.
2. Talk or sing: you hear the internal chord speaking. Turn **Start** for other chords, **Pitch** (or V/oct) to play them, **Density** for a wider supersaw.
3. Press **Freeze** on a vowel and play chords on it. Press it again to let go.
4. Turn **Speed** past a click to shift the formants. **Size** fully clockwise for a sharper robot.
5. Press **Record** (Track) and sing: the chord follows your pitch. Or press **Source** and patch a synth into the right input as the carrier.

Notes

- The bands run from 100 Hz to 8 kHz, a third of an octave apart, each a 4-pole band-pass.
- The output level follows the modulator's level and the carrier's brightness: a higher chord root sounds a little louder, and an external carrier works like a VCA input (its level matters).
- The voice on the left input isn't delayed: Blend mixes it back exactly.
- CPU, measured offline: about the same as shimmer. The pitch tracker only runs while Track is on.

stretch — extreme time-stretch and freeze

A Csound instrument that slows sound right down, up to 1000× slower, **without changing its pitch**. It can also freeze a moment and hold it, play backwards, or scrub through a sound by hand. The sound is either a file from the stick, or what you just played into the inputs: Record grabs the last few seconds, so you never miss the moment. Layers, smear, blur and a reverb turn a few seconds of anything into a slowly breathing landscape.

It's in the **Csound user bank** as `006_stretch`. Under the hood is a phase vocoder (`mincer`) with a time pointer that the panel moves. The pointer can stand still, crawl, run backwards or jump, and the pitch stays independent of it.

Panel map

```
          [ speed ]              speed   = STRETCH  centre frozen, CW fwd, CCW back
start                   size     start   = POSITION (turning it scrubs)
                                 size    = REGION the playhead loops in
density    [ pitch ]   overlap   density = LAYERS 1-4, offset in time
                                 window  = SMEAR  clear -> washy
     blend      window           overlap = BLUR   smoothed over 0-2 s
                                 blend   = dry <-> stretched
                                 pitch   = TRANSPOSE, 1V/oct

[record]  [file]    [source]   [reset]    [freeze]
 GRAB      NEXT      FILE/LIVE  TO START   PAUSE
```

Controls

Control	Does
Speed	Centre = frozen in place. Turn clockwise to move forward: glacial at first, reaching the original speed at the click. Past the click it goes faster, up to 4×. Anticlockwise does the same backwards. The rate curve is logarithmic, so most of the knob is the slow end. A gentle forward crawl is set when it loads
Start	Where in the sound the playhead is. Turning it scrubs the playhead through the sound, like moving a tape by hand
Size	The region the playhead travels in, starting at Start. It wraps round at the end. Small = a short loop of one moment; fully clockwise = from Start to the end of the sound
Pitch + 1 V/oct	Transpose, ±3 octaves; the click = original pitch. Independent of Speed: a frozen sound can be played as a note
Density	Layers: fully anticlockwise = one reader. Turning up fades in three more, each reading a little earlier or later in the sound and placed across the stereo field. Gives a thick, chorus-like cloud of the same moment
Window	Smear. Anticlockwise = clear and precise. Turning up reads each analysis frame from a slightly different place, so the sound turns washy and diffuse (paulstretch-like). Past 70 % the phase locking is also released, for the full wash
Overlap	Blur. The spectrum is smoothed over time, from none up to 2 s: changes melt into each other and sounds hang on
Blend	Dry input ↔ stretched sound (equal power). Fully clockwise = only the stretch

Button / jack	Does	LED
Freeze (button or gate)	Pause: holds the current moment, whatever Speed says. Press again to carry on. A gate toggles it	on = paused
Reset (button or gate)	Jump back to Start, with a short soft dip. Clock it to re-strike in time	—

Button / jack	Does	LED
Record (button or gate)	Grab the last N seconds of the input (N = alt Size, default 30 s). It is always listening, so press it *after* something good happened. The grab plays when Source is on	—
File	Next file from the stick	—
Source	Off = stick files. On = the grabbed input. With nothing grabbed yet, the first 5 s you play are grabbed by themselves	on = live

Alt layer (hold Source)

Control	Does	Default
alt **Start**	Layer spread: how far apart the layers sit in the stereo field	half
alt **Size**	Grab length N: 5 – 120 s	30 s
alt **Density**	How far apart in time the layers read: 0.02 – 4 s	≈ 0.4 s
alt **Window**	Tone: dark (−18 dB treble) ↔ bright (+6 dB); noon = flat	flat
alt **Overlap**	Reverb size	medium
alt **Blend**	Reverb amount	a little
alt **Pitch**	Layer detune: up to ±30 cents between the layers (turn it past noon)	none
alt **Speed**	Not used (it's the Nebulae's input gain)	—

Quick start

1. Put a few audio files on the stick (anything: a voice, a chord, a field recording). Load `006_stretch`. It starts crawling through the first file. Set Blend fully clockwise.
2. Turn Speed towards the centre: slower and slower, until it freezes. Turn Start to scrub to another moment.
3. Turn Pitch: the frozen moment becomes a note. Patch V/oct and play it.
4. Bring up Density for layers, Window for smear, Overlap for blur. Hold Source and raise Blend for reverb.
5. Live: play something into the inputs, press Record, tap Source (LED on). What you just played is now frozen in time.

Notes

- Long files and long grabs cost memory, not CPU: up to 120 s of input is kept twice (one buffer recording while the other plays).
- CPU with everything on is measured at about 1.2–1.5× the factory granular looper (offline, PC).
- Files at any sample rate play at the right pitch and speed.
- Reset and File changes dip the sound for a moment (20 ms down, 40 ms up), so jumps don't click.

bt_storyteller: the Nebulae tells your PC what is happening to it

A proof of concept. The sound is the factory **granular looper**, unchanged. Behind it, the Nebulae's Pi 4 switches on its own Bluetooth radio, pairs with your PC as a **Bluetooth keyboard**, and types what is happening to it: knob moves, CV, buttons, pulses, sound and silence. Each observation comes as a line of raw data, then a line of poetry. It proves the Nebulae can talk Bluetooth with nothing installed on the SD card, while an instrument runs at full quality. Everything it needs comes from the USB stick and runs from RAM. Bluetooth is only on while this instrument is loaded. The engineering is in `docs/BLUETOOTH.md`.

1. On the stick

Where	What
Stick root	`bt_storyteller.instr` (the instrument; Csound user bank)
`bt_keyboard/`	`btkbd` (the keyboard program), `run.sh` (starts it), `narrator.py` (the storyteller), `phrases.py` (its voice), `settings.txt` (data lines, pace), `README.txt`
Written by the module	`pairing.tlv` (your PC), `narrator.log` (everything it typed), `log.txt` (the Bluetooth side), and the previous three runs of each as `.1`, `.2`, `.3`

2. First time: pairing

1. Load **bt_storyteller**: hold Speed, Speed anticlockwise (Csound user bank), then find it with the Pitch pages (see the bank sheet, `nebulae_banks.txt` on the stick).
2. The looper plays as usual. Every few seconds you hear a **soft double blip**: it is waiting to pair.
3. On the PC: **Settings → Bluetooth & devices → Add device → Bluetooth → "Nebulae"**. No PIN is asked.
4. A **rising chime**: connected. The pairing is saved to the stick (`pairing.tlv`).
5. Open **Notepad** and click in it. **It types into whichever window has the focus**, so keep Notepad in front (and nothing important, like a terminal or a chat box).

After that it reconnects to the same PC by itself each time you load it. It is not discoverable once it has a PC, so nobody else can pair with it.

A **low tone** means it failed. The looper keeps playing; `bt_keyboard/log.txt` says why.

To pair with a different PC: delete `pairing.tlv` from the stick and remove "Nebulae" from the old PC's Bluetooth devices.

3. Playing it

The panel is the factory granular looper (see Qu-Bit's manual). Every control is also an input to the storyteller:

It notices	How
Knobs	each move: from → to, how long it took, reversals. Indecision (back and forth), small, big, to the top or bottom, back to an earlier setting, the first touch of each knob, several at once. Pitch moves as intervals (an octave, a fifth); Speed through backwards, stopped, normal
CV	a knob moving by itself: a wave (with its period), steps (a sequence) or wandering
Buttons and gates	Freeze, Record, Source, File (next file), the hidden layer (Source held), Reset as one press or a burst
Pulses	on the Reset gate and in the input: regular, loose or irregular, the tempo, faster, slower, stopped
Sound	a sudden transient after calm, silence (20 s, 60 s, then every 3 min), sound arriving, brighter or darker, more or less low end, louder or softer, its own voice louder than the input, frozen and alone
Memory	a sound it heard over 45 s ago coming back: "I recognise this sound", and what is different now

It notices	How
Time	musings when nobody touches it; how long you have been together

Things to try: turn a knob slowly, then quickly, then back and forth. Patch an LFO and then a sequencer into a CV input. Clock Reset regularly, then irregularly. Play into it, stop for a minute, then bring the first sound back.

4. Reading it

```
[1:39 heard 94 seconds ago: in -32 > -26 dB, 231 > 152 Hz, lows 78% > 93% / in -26 dB, out -33 dB, 153 Hz, lows 93%, calm]
I know this. I heard it 94 seconds ago. There is more weight underneath. The edges have softened.
```

- **The data line**, in brackets: the time since the instrument loaded (m:ss), then what it noticed. After the `/` is the sound at that moment: input level, output level, brightness (the spectral centroid, in Hz), the share of low end (below 160 Hz), and the mood (calm or busy, from how much has been touched in the last 30 s).
- **The poem**: a line from its voice (`phrases.py`), sometimes with a closing musing (a "coda").

It writes one observation at a time, at about 26 characters a second, and drops anything that has gone stale (after 12 s) rather than fall behind.

5. Settings: how much data, how chatty

Edit `bt_keyboard/settings.txt` on the PC (Notepad is fine), put the stick back and reload the instrument:

Line	Values	Default
`data =`	`full`: time, what happened and the sound, as above. `short`: time and what happened only, e.g. `[1:05 density 0.42 > 0.71 in 0.8 s]`. `off`: poetry only	`full`
`pace =`	seconds between observations: `1` = chatty, `10` = thoughtful, `30` = taciturn (urgent ones can still come after 1 s)	`3.5`
`coda =`	percent of lines with a closing musing, `0` … `100`	`20`

A misspelt line or value is ignored and keeps its default. The first line of `narrator.log` shows the settings it used. (An empty file named `hide_data` in `bt_keyboard/` still works too: it is the same as `data = off`.)

6. Changing its voice: `phrases.py`

Everything it can say is in `bt_keyboard/phrases.py`: 343 lines in 94 pools. Open it in a text editor (Notepad, or VS Code). Each pool is a list of lines in double quotes, separated by commas:

```
TRANSIENT = ["Something has interrupted me.", "A shock. What was that?", ...]
```

- **Add, change or delete lines freely.** More lines = fewer repeats: it deals each pool like a shuffled deck, so nothing repeats until a pool is used up.
- **Fields** in `{braces}` are filled in: `{name}` / `{Name}` (the control's poetic name, from `NAMES`; capital = sentence start), `{dur}`, `{n}`, `{bpm}`, `{period}`, `{interval}`, `{dir}`, `{ago}`, `{diff}`, `{what}`, `{mins}`. Use only the fields a pool already uses (the comment above each pool says what it is for).
- `"\n"` inside a line breaks it into a verse.
- **Characters:** only letters, digits, space and `. , ! ? ( ) - : % ' / = + < > * ; [ ]`. These type the same on UK and US keyboards. Anything else (quotes inside a line, `@ # ~ | & _ ^`, accents) is typed as a space.
- Keep the pool names (the CAPITALS) and the brackets as they are.

- Some pools are worth knowing: `NAMES` (how it names each knob), `UP` / `DOWN` (per knob), `GREET` (its first words), `CODA` (the closing musings), `IDLE_SOUND` / `IDLE_SILENT` (when nobody touches it), `RECOGNISE` and `DIFF` (memory).

If you break it, the looper still plays, but it gives the low tone and types nothing; `log.txt` says "narrator.py or phrases.py does not compile". Usually a missing comma or quote. To check on the PC first: in the repo, `python bt/tests/test_narrator.py` checks every line, or at least `python -m py_compile phrases.py`.

7. Logs and what to bring back

- `narrator.log`: every observation it made, with times. `log.txt`: the Bluetooth side (pairing, connecting, shutting down). The three runs before are kept as `narrator.log.1` … `.3` and `log.txt.1` … `.3`, so a short test load doesn't wipe a good session.
- For tuning, bring back those files plus what felt wrong: too chatty, too slow, observations that ring false, lines that get tired.

8. Limits

- A keyboard can only type, so the PC never talks back. It can't read a reply.
- UK/US-safe characters only (no quotes or symbols).
- Tested with Windows 10 and a UK layout.
- The clock counts from when the instrument loaded (the module has no real-time clock).
- It runs only while `bt_storyteller` is loaded. Loading another instrument disconnects the PC and turns the radio off within a few seconds.
- CPU: the listening adds about 7 % to the looper, so the looper itself is unaffected.
- `bt_test` (the stage 1–2 check: pairing plus a hello line every 10 s) is still on the stick for diagnosing Bluetooth problems.

Utilities on the stick

These aren't instruments. Each one does one job, then lets audio pass (or plays a test sound).

File	Bank	What it does	What you hear / get
999_fw_backup	Csound user	Backs up the module's own firmware to `neb_fwbackup/` on the stick (the revert package, first run only, never overwritten), and writes an inventory: OS, versions, packages, boot files, the hash of every firmware file. Changes nothing on the module	Audio passes through. Then every 4 s: **three high beeps** = done and verified; **one long low tone** = something failed (see `inventory_*.txt`)
bt_test	Csound user	The Bluetooth check: tests the radio chip's power and serial line (`diag.txt`), starts the Bluetooth stack (`probe.txt`), then acts as a keyboard that types "Hello from the Nebulae" every 10 s once paired	Blips while waiting to pair, a chime when connected, a low tone on failure. Files in `bt_keyboard/`
ZZ_pitchlog	Pd	Measures how the firmware delivers Pitch: Reset starts a 20 s capture of every Pitch message. Run `999_fw_backup` afterwards to copy the capture to the stick	Nothing special. Used for the timing measurements
01_sctest	SuperCollider	Checks that SuperCollider runs and every control arrives: a plucked saw	Reset plays a note

Safe to keep on the stick. They only run when you load them.